Azure Regional Disaster Recovery

Day 06 | AZURE / AKS + POSTGRESQL + STORAGE

Use this guide to investigate a specific failure, apply a cause-specific change and verify recovery. Start with evidence from the affected workload. The examples require adaptation to your environment.

How the failure happens

Regional recovery crosses routing, compute, identity, networking and state. A healthy standby HTTP listener is insufficient when the database remains read-only or a storage write endpoint is unavailable. Recovery orchestration must make the whole application ready.

The Scenario Front Door reaches standby AKS. PostgreSQL is still a read replica and Storage writes are unavailable.

Why Traffic routing does not perform database promotion or account failover.

Prepare First Deploy standby capacity, replication, regional access and a tested recovery workflow.

Before you start

Defined RTO/RPO; standby AKS deployment and capacity; a PostgreSQL cross-region replica; eligible geo-redundant Storage; secondary access paths; a recovery coordinator outside the primary region.

Diagnostic checks

```
# Check primary + standby app readiness
# Inspect PostgreSQL replica lag and role
# Inspect Storage replication + last sync
```

Define The Targets: RTO is acceptable downtime. RPO is acceptable data loss.

Check Independent Signals: Do not force data failover from one failed HTTP probe.

Apply the cause specific fix

Fix The Confirmed Cause

```
# DR orchestration blueprint, NOT runnable API
steps:
- confirm-regional-failure
- fence-old-writer
- promote-postgresql-replica
- recover-required-storage-writes
- reconnect-and-test-standby
- enable-standby-application-readiness
```

Automate With Guardrails: Run orchestration outside the failed region, with scoped identity, retries, state tracking and a single coordinator.

Data Loss Is Possible: Cross-region replication is asynchronous. Forced promotion/account failover can lose recent writes.

Validate the result

```
# Test from the standby application:
# 1. Database write + read-back
# 2. Required Storage operation
# 3. Complete business request
```

Expected: Front Door sends traffic only to an application ready to use its dependencies.

After Recovery: Rebuild protection and plan failback. Do not immediately switch writers back.

Practice exercise

Rehearse failover with disposable data and record detection, promotion, reconnection and validation timings. Test a write plus read-back, and confirm the old writer cannot independently resume.

Implementation notes

Blueprint only; no automatic destructive operation is supplied. PostgreSQL Flexible Server cross-region promotion is customer-initiated and can be orchestrated; use writer virtual endpoints with the appropriate promote-to-primary operation. Fence writers to prevent conflicting updates. Storage geo-secondary location follows Azure configuration/pairing and may differ from chosen app DR region. RA-GRS secondary is read-only until write failover. Supported account features vary; private DNS and endpoints must work after recovery. Database and blob replication are not transactionally coordinated.

Interview question

Does Front Door promote PostgreSQL or fail over Storage? No. It routes traffic to origins; data-service recovery needs its own supported operations and safeguards.

Record what changed

- The symptom and affected user operation.
- Evidence supporting the root cause and alternatives ruled out.
- The changed configuration and the version or identity involved.
- Positive and negative tests, plus any remaining risks.

Example status

Examples are educational templates or fragments. They were not deployed against a live cloud account or production cluster. A successful syntax check is not proof of runtime compatibility. Test adapted configurations in an isolated environment first.

Official references

Official documentation 1

Official documentation 2

Official documentation 3